

Managing Technological Innovation: Dynamic Capabilities, Collaborative Innovation, and Born-Digital SMEs' Performance

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Abstract—In contrast to the digital transformation of the incumbents, born-digital small- and medium-sized enterprises (SMEs) embrace digital logic in the organization's design, processes, and governance from the very start. So far, scholars have focused on the digital transformation of the incumbents; thus, little is known about how dynamic capabilities, collaborative innovation, and firm performance interact in the context of born-digital SMEs—a new generation of digital firms with no legacy of the incumbent firm. Therefore, the study's research question is how dynamic capabilities and collaborative innovation affect the performance of born-digital SMEs. We aim to investigate the impact of the dynamic capabilities of born-digital SMEs on their performance when collaborative innovations mediate it. The empirical research was based on 268 complete viable responses from born-digital SMEs in the U.S., Sweden, and Lithuania. The study results show that dynamic capabilities' effect on born-digital SMEs' performance is more potent when mediated by collaborative innovations. Furthermore, the study highlights that although the impact of dynamic capabilities on partnership networks and relational collaboration is significant, only relational collaboration significantly affects the performance of born-digital SMEs. Our study contributes to the management of technology literature by advancing knowledge on dynamic capabilities and collaborative innovation of born-digital SMEs. In addition, we are exposing the limited role of partnership networks in born-digital SMEs' superior performance and spotlighting the significance of relational collaboration—cocreating innovative products and services and sharing the cocreated value among the relevant partners.

Index Terms—Born-digital, collaborative innovation, dynamic capabilities, performance, relational collaboration, small- and medium-sized enterprise (SME).

I. INTRODUCTION

THE management of technological innovation is critical for the success of small- and medium-sized enterprises (SMEs) in today's volatile, uncertain, complex, and ambiguous

(VUCA) business environment [1]. Digital technologies have shaped markets and industries for the past decade, introducing new possibilities and challenges [2]. As a result, a new generation of organizations called “born-digital” have emerged that rely on digital logic and infrastructure from the very start of their existence. The emergence of born-digital SMEs has been a major disruptor in the technological innovation landscape [3]. We define “born-digital” as firms solely based on the digital value chain and business model empowered by the digital technology infrastructure [4]. These firms embrace digital technologies and connected network logic as their core competence, relying on collaborative innovation in different forms from the very beginning, which manifests as their ability to develop new products and services quickly, create innovative business models, and adapt to changing market conditions [5], [6]. Such more known firms are, e.g., Uber, Google, and Amazon. However, many born-digital new entrances have yet to be discovered but significantly impact the economy and social well-being, e.g., Omada Health, Lift, and Movendos. Although the number of born-digital SMEs is rapidly growing, inspired by the dazzling success story of GAFAM,¹ most born-digital SMEs fail due to network externalities, lack of continuous, long-term investments, insufficient capabilities, and other relevant factors [7]. Born-digital SMEs are unique in comparison with traditional and non-born-digital SMEs in three ways: 1) core competence is based on digital technologies from the very inception; 2) they are flexible and rely on complements when advancing value proposition; and 3) they are not sensitive to business sectors and geographical boundaries and, thus, scalable [4], [8]. These unique features of the born-digital SMEs call to revisit how they build and maintain their competitive advantage [9]. Moreover, for most born-digital SMEs, competing only based on resource superiority or availability is challenging due to the digital business's modular, connected, and scalable nature [10], [11], [12]. They constantly need to integrate, realign, and reconfigure these resources to support ever evolving and changing architecture of the born-digital SME product and services (e.g., software as a service (SaaS), platform as a service (PaaS), infrastructure as a service (IaaS) and application software).

By definition, dynamic capabilities create a sustainable competitive advantage for a firm in an increasingly volatile environment [13], [14], [15], [16]. Thus, they aid born-digital

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¹GAFAM—Google, Amazon, Facebook, Apple, and Microsoft

firms to sense, seize, and transform new external and internal digital business opportunities by integrating, reconfiguring, and transforming a firm's core competencies following internal and external resources and assets [17]. Furthermore, to access external resources and assets, born-digital SMEs engage in collaborative innovation through partnership networks and relational collaboration [18]. Collaborative innovation is defined as innovation activities carried out through the cooperation of several different partners with a collective purpose and expected innovation outcome [19], [20], [21]. Although dynamic capabilities and collaborative innovations are well explained concerning technological incumbents and their digital transformation [22], [23], [24], born-digital SMEs lack an understanding of how they can utilize dynamic capabilities and collaborative innovations for their benefit. So far, little is said about how dynamic capabilities and collaborative innovations enable born-digital SMEs to instantly access rapidly changing knowledge of digital technology management and sustain competitive advantage in multisided markets [4], [25].

Prior management of technology (MoT) research reports that interaction between dynamic capabilities and core competence is the primary source of sustainable competitive advantage, which is consequently measured by a firm's performance indicators [26], [27], [28]. In contrast to RBV, both dynamic capabilities and competence-based perspectives contend the need to constantly and dynamically develop—integrate, reconfigure, and transform—relevant capabilities and/or competencies [29], [30]. The competency-based perspective suggests that core competence can be defined as collective learning, linking individual assets within the firm that manifest in increased customer satisfaction and overall product/service success [30], [31], [29]. According to Teece [32], dynamic capabilities determine and shape of the firm's core competence. Born-digital firms have a more remarkable ability to outsource and utilize any readily available digital technologies, thus, traditional competitive edge sources that would help stand out from the competition are no longer applicable. Furthermore, digital technologies degrade borders between industry sectors, including industry-specific core competence, offering connectedness instead as a core competence of the born-digital SMEs [24], [26], [29], [33]. Therefore, new insights and knowledge on how born-digital SMEs can utilize their dynamic capabilities and core competencies are needed [16], [17].

Furthermore, as digital technology becomes increasingly connected and complex, SMEs immerse themselves in exploiting external sources to access complementary knowledge and new markets, share costs and risks, and enhance innovation performance [18], [19], [20]. This can be accomplished through open innovation [21] and collaborative innovations. However, researchers are concentrating on some distinct areas of the open innovation framework [35]; as a result, the literature on the open innovation concept in SMEs is scarce and fragmented [12], [36]. Areas such as collaborative innovation in SMEs have been developed sparingly. Prior research on collaborative innovation focused on the interaction between different firms and the stakeholders [37], [38], [39]. Organizations increasingly engage in collaborative innovation and do so in various ways,

for example, with partners, suppliers, customers, and users, by bridging organization boundaries to tap into knowledge from different sources and by cocreating solutions [24], [35], [40], [41]. As a result, collaborative innovation grants access to an external knowledge base and assures joint efforts to develop new products and services.

At this point, dynamic capabilities can aid the integration, realignment, and reconfiguration of internal and external assets (e.g., core competence, new external knowledge, and capabilities) to produce new digital products and services. Dynamic capabilities are proven to be shaped by context [42].

Therefore, the research question of our article is: how do dynamic capabilities and collaborative innovation affect the performance of born-digital SMEs? Our study aims to investigate dynamic capabilities' impact on born-digital SME performance mediated by collaborative innovations. The specific dynamic capabilities depend on the content and context of the discovery process. Thus, empirical research is needed to investigate the role of dynamic capabilities and collaborative innovations in born-digital SMEs' performance.

Based on the literature analysis, we argue that collaborative innovation significantly mediates the relationship between dynamic capabilities and born-digital SMEs' performance. Additionally, we delineated our study in the contexts of born-digital SMEs operating in the countries established as innovation leaders (USA and Sweden) and moderate innovators (Lithuania) [43]. We investigated born-digital SME operations in three business sectors: information and communications technology, health services and medicine, and financial services.

The study proposes a threefold theoretical contribution. First, we proved that dynamic capabilities are the antecedents of collaborative innovations in the case of born-digital SMEs. We confirm that the dynamic capabilities of born-digital SMEs positively affect collaborative innovation. In other words, the stronger the born-digital SMEs' dynamic capabilities are, the more advanced collaborative innovation is. Second, in contrast to earlier studies, we have confirmed that partnership networks are insufficient to assure superior performance of born-digital SMEs if the partnerships are not based on relational collaboration. Third, dynamic capabilities are insufficient to assure superior performance for born-digital SMEs. Relational collaboration is vital to foster superior performance and competitive advantage of born-digital SMEs'. In other words, collaborative innovation through relational collaboration significantly mediates the relationship between dynamic capabilities and born-digital SMEs' performance.

Our article provides empirical evidence that the dynamic capabilities of born-digital SMEs ensure advanced collaborative innovation by supporting partnership networks and relational collaboration. The study's findings are relevant to the born-digital SMEs as they should be mindful of the organizational processes—dynamic capabilities—that are significant while embracing and developing digital technologies through collaborative innovation approaches.

The rest of this article is structured as follows. First, we define the main concepts of the study and provide theoretical background and hypotheses in Section II. Literature analysis

juxtaposes main ideas on born-digital SME performance, collaborative innovation, and dynamic capabilities. Subsequently, we display our research methodology and sample characteristics in Section III. Section IV offers our research structural results and tests the hypotheses. Finally, Section V lays out the research's discussion, implications, and conclusions.

II. THEORETICAL FRAMEWORK

A. Defining Born-Digital SMEs

Our study focuses on born-digital SMEs and how dynamic capabilities and collaborative innovation affect their performance. Born-digital SMEs represent a unique research context as they are a relatively new type of firm that have emerged in the digital age. Unlike traditional SMEs, which may have started before the advent of digital technologies, born-digital SMEs have embraced digital technologies from their inception, and their business models are inherently digital. However, born-digital SMEs still need to ensure their competitiveness in the VUCA environment. Thus, born-digital SMEs must invest in the resources and capabilities necessary to compete with incumbents and other firms in their industry [44]. Consequently, SMEs create value and achieve superior performance by integrating and reconfiguring interorganizational networks as an external source of complementary assets and integrating them into their business models [45]. Furthermore, dynamic capabilities aid SMEs in creating, integrating, and reconfiguring internal and external knowledge, often generated through collaborative innovations [46]. This is especially true because born-digital SMEs rely on complementors and networks when advancing their value proposition and scaling [4], [8]. Next, we specify the unique features of digital-born SMEs that can make change in how dynamic capabilities, collaborative innovation, and performance manifest and interact. It is essential to understand how these prevailing born-digital SMEs build and maintain their competitive advantage.

Many of these born-digital SMEs embrace operating exclusively digital and position this digital business model as their core competence [23], which offers more scalability and flexibility options. This distinguishes them significantly from ordinary physical firms (bricks-and-mortar), which started the digitalization process sometime after their inception and, as previously indicated, cannot be considered as born-digital firms. Notably, born-digital firms are characterized by operating online, thanks to what they have immediate access to all markets worldwide.

When comparing traditional large multinationals, digital firms stand out in the way they rely on digital technologies and infrastructure to communicate and collaborate. Vadana et al. [4] also argue that while there are similarities between born-digital and born-global firms, it must be noted that not all born-digitals are internationalized, and not all international firms are born-digital.

In earlier literature, concepts such as "Internet-based business" [47], "pure-Internet firms" [48], and "e-commerce corporations" [49] have been documented and analyzed, primarily considering their emergence due to the rapid adoption of Internet technologies in business and households. "Internet-based" business is defined as firms participating in online communication,

business proposal, and Internet-enabled transactions as the basis for their business model [50]. We view the Internet-based business concept as the predecessor of born-digitals, which in contrast, emerged later and is defined through more robust business models and a broader scope of digital technologies, not only those used by Internet-based or Web 3.0 firms [51]. Born-digital SMEs' business models are empowered by the availability and affordability of digital technologies, such as cloud computing, data servers, 5G, IoT, AI, VR, etc., are dealing with data as a core asset, and are multisided and, thus, need to master complex configuration of multiple sides. This is a radical change compared to Internet-based businesses and needs further investigation of born-digital dynamic capabilities, collaborative innovation, and their performance, e.g., [18], [34], [47].

Summing up, we analyze born-digital SMEs within this research, defined by three main facets. First, these SMEs rely solely on digital products and services (e.g., PaaS, SaaS, IaaS, and application software), operating on and leveraging digital infrastructure (e.g., cloud computing, digital data centers, etc.) [6], [33]. Second, they operate on digital business models and create online business offerings [6], [33]. Finally, from their inceptual foundation, they fully satisfy the first two criteria [4], [6].

From the very beginning, born-digital SMEs focus and invest in their core competence, which is being digital-first. Thus, the ability of born-digital SMEs to develop and leverage dynamic capabilities is crucial for their success [32], [52], [53]. However, to translate these capabilities into improved performance, born-digital SMEs might need collaborative innovation [9], [46]. Partnership networks and relational collaboration are significant for born-digital SMEs because these firms operate in rapidly evolving and highly competitive industries [54]. By adapting quickly to changes in the VUCA business environment, these firms can maintain a competitive advantage and continue to innovate. In addition, partnership networks and relational collaboration enable born-digital SMEs to access the resources, knowledge, and capabilities necessary to develop dynamic capabilities, adapt, and shape their business environment effectively [4].

Table I summarizes the literature review, indicating a need to investigate further interrelationships between dynamic capabilities, collaborative innovation, and born-digital SMEs performance. Furthermore, Table I lists a clear gap in understanding the impact of collaborative innovations, particularly partnership networks and relational collaboration, on born-digital SMEs.

For the reasons discussed above, in the following sections, we investigate more in depth how dynamic capabilities and collaborative innovation enhance the performance of born-digital SMEs with no legacy of incumbent firms.

B. Dynamic Capabilities and Born-Digital SMEs' Performance

The dynamic capabilities framework was developed based on the studies of multinational firms [55], explaining how they remain competitive in VUCA business environments stimulated by technological innovations. Teece [56] notes that dynamic

TABLE I
SUMMARY OF REVIEWED MAIN LITERATURE

Dynamic capabilities and collaborative innovation need further analysis in the context of born-digital SMEs	Dynamic capabilities are essential for digital firm's performance	The effect of dynamic capabilities on a firm's performance is indirect and depends on context	Collaborative innovation is a mediator between dynamic capabilities and firm performance	Partnership networks have a significant impact on born-digital SMEs performance		Relational collaboration has a significant impact on born-digital SMEs performance
Brunswick and Vrande [12] Usman et al. [36]	Teece [55] Teece [56]	Helfat and Peteraf [13] Zahra et al. [15]	Chesbrough [21] Protogerou et al. [57]	Vadana et al. [4]	Monaghan et al. [6] Johanson and Vahlne [58] Birkinshaw [60]	Vadana et al. [4]
Caccamo [37]	Helfat and Raubitschek [52] Huang et al. [61]	Ferreira et al. [28]	Pundziene et al. [59]			
Radziwon and Bogers [24] Teece [64] Teece [55]	Teece [22] Boscoianu et al. [53]	Bernroider et al. [65] Eisenhardt and Martin [67] Protogerou et al. [57] Santoro et al. [71]	Bogers et al. [63] Zhang et al. [66] Lopes and de Carvalho [68]			
Wilden et al. [69] Westerman et al. [70] Schilke et al. [62] Gamra et al. [73]		Wilden et al. [72] Karimi and Walter [74]				
Arias-Pérez [75]						

capabilities are the antecedents of firm scope, the potential for growth, sustained profitability, and performance while operating in VUCA markets. Later research [69] has suggested that the dynamic capabilities framework can help incorporate necessary processes to achieve superior performance for various organizations. With the emergence of born-digital SMEs, dynamic capabilities are needed to detect, select, and integrate external knowledge, move fast and be agile, and constantly innovate and reconfigure internal and external resources to be ready for new business opportunities [70].

The dynamic capabilities framework explains how the organizational and core competencies of the firm are transformed over time into a competitive advantage and superior performance in open and technologically driven economies [55]. Moreover, various studies have identified dynamic capabilities as an essential factor in sustaining the firm's superior performance [55], [62], [69]. Combined with the potential of the digital firm, born-digital SME dynamic capabilities may unlock an unprecedented pace of business scaling [61]. The dynamic capabilities framework helps identify factors likely to impact firm performance [22]. In the context of the development and adoption of digital technology, dynamic capabilities are required to explore the momentum and dynamics of the opportunities, achieve successful collaboration with diverse partners inside and outside the organization, and in that way support digital business scaling [53], [56].

The specific dynamic capabilities depend on the content and context of the discovery process [26]. For example, big technology firms, such as Google, Amazon, Facebook, and others, have strong digital platform building and Big Data analytics capabilities, in contrast to manufacturing firms, such as GE, Philips, Continental, and others, with strong industry domain capabilities. Thus, we argue that born-digital SMEs are also a specific context that generates unique dynamic capabilities and their effect on firm performance.

Different scholars have reported diverse positions regarding dynamic capabilities and firm performance. Schilke et al. [62] have identified at least 113 studies considering primarily the positive effects of dynamic capabilities on firms' performance. Other scholars [57], [65], [67], [71], [72] have suggested that the effect of dynamic capabilities on a firm's performance is indirect and mediated by various factors, such as operational processes and capabilities (e.g., open innovation). However, it is still not fully documented how dynamic capabilities affect the performance of born-digital SMEs, as most dynamic capability studies remain focused on large incumbent firms. While much is said about what and how dynamic capabilities enable organizations to be timely and effectually respond to the rapidly changing technology-intensive business environment, little is known about how dynamic capabilities manifest in the context of born-digital SMEs.

Thus, following previous discussions of the scholars leads us to the development of the first hypothesis.

H1. Dynamic capabilities have a significant direct positive effect on born-digital SMEs' performance.

C. Dynamic Capabilities and Collaborative Innovation

As mentioned above, Eisenhardt and Martin [67] and Protogerou et al. [57] questioned the direct positive relationship between dynamic capabilities and firm performance. Instead, they reckoned that the relationship is not straightforward but somewhat mediated by firms' processes, recourse base, and capability configurations. This has been expanded in later studies [13], [15], [62], [74], [76] showing that mediating factors help better explain the effects of dynamic capabilities on firm performance. Innovation processes, in particular, open innovation, [63] are possible mediating factors between dynamic capabilities and a firm's performance, which embody purposeful knowledge flows

across organizational boundaries among internal and external networks [21], [59], [63].

For some time, discussions about open innovation and dynamic capabilities have revolved around their external and internal perspectives [35], [77], [78]. Collaborative innovation as part of open innovation has been researched following the resource-based theory, where the way firms complement their assets and try to reduce innovation costs was analyzed [79], [80]. Bogers et al. [63] recognized that in today's digital-based environment, sensing, seizing, and orchestrating skills are inherent in the collaborative innovation field.

We define collaborative innovation as an open innovation strategy that helps create organizational and business improvements by intertwining and balancing internal and external knowledge and relationships of the actors [37]. Collaborative innovation is a broad term. However, it is possible to identify two significant components: partnership networks [38] when multiple actors engage in a product, services, policies, and business innovation activities; and cultivation of mutual relationships (which we call relational collaboration) that cannot be generated by either firm in isolation and can only be generated through the idiosyncratic cocreation and sharing the benefits, e.g., trust, reciprocity, and goodwill [81]. However, the existing literature still lacks observations of the role of dynamic capabilities in the collaborative innovations [73], especially concerning the born-digital SMEs context.

Our research delves into the born-digital SMEs that do not require digital transformation. Thus, dynamic capabilities are needed not for transformation but to quickly acquire relevant knowledge and resources through the collaborative innovation [66]. In developing and utilizing digital technology, dynamic capabilities are required to explore the momentum and dynamics of the opportunities to achieve successful collaboration with diverse partners both inside and outside the organization [53], [56]. Dynamic capabilities are important for born-digital SMEs as they enable these firms to adapt and respond to changes in the business environment [19]. However, to translate these capabilities into improved performance, born-digital SMEs must be able to leverage them effectively to create innovative products and services that meet the needs of their customers. Collaborative innovation can enable born-digital SMEs to leverage their dynamic capabilities effectively by providing access to external resources, knowledge, and expertise [59].

Kohler et al. [81] combined dynamic capabilities, open innovation, and relational view, which further advanced the understanding of collaborative innovation as a composite of not only partnership networks but also as relational collaboration—"a supernormal profit jointly generated in an exchange relationship that cannot be generated by either firm in isolation and can only be generated through the idiosyncratic joint contributions of the specific alliance partners" [82, p. 662]. Thus, dynamic capabilities can help to orchestrate partnership networks [57], [63] and cultivate inter- and intra- firm relational collaboration through the transformation of organizational culture and relational patterns [19]. Further, in more detail, we review partnership networks and relational collaboration as constituent parts of collaborative innovation.

First, partnership networks are defined as extensive interorganizational collaboration between the partners and other collaborators needed to run the business (e.g., suppliers, knowledge partners, etc.) [83]. It is important to spotlight that partnership networks, in contrast to relational collaboration, can be delineated as "loosely-coupled systems, where the participants are neither linked by capital nor through joint efforts in a specific project or business areas, but rather by more general agreements" [84, p. 250]. These, in turn, provide benefits, opportunities, and better access to resources, which improves the competitive advantage of a firm [20], [35], [40], [41], [83]. Furthermore, partnership networks are a natural and significant part of businesses empowered by digital technologies such as, for example, platforms. Most born-digital SMEs depend on the network effect; thus, their performance is affected by the number and type of partners they engage and maintain [4]. By establishing partnership networks and engaging in relational collaboration, born-digital SMEs can access complementors resources, knowledge, and capabilities necessary to innovate and compete effectively.

Second, we define relational collaboration through the relational view perspective suggesting that benefits and value are cocreated and jointly owned by partnering stakeholders [20], [82], [85], [86], [87]. Relational collaboration rests on the breadth and depth of the interpersonal relationships network members develop with each other, and such relational collaboration capability significantly impacts firms' product innovation [88], [89]. Value cocreation and sharing the benefits with partners and customers allow born-digital firms to fully use the partnership networks and assets across the respective value-chain [4]. By engaging in relational collaboration with their partners and complementors, born-digital SMEs can access tacit knowledge, build trust, and develop long-term relationships that can be leveraged to create innovative products and services.

While much is said about how dynamic capabilities enable organizations to respond quickly and effectively to the rapidly changing business environment, little is known about how dynamic capabilities affect partnership networks and the relational collaboration of born-digital SMEs. Given the potential of dynamic capabilities on partnership networks and relational collaboration, we set forth the following two hypotheses.

H2a. Dynamic capabilities have a significant positive effect on partnership networks of born-digital SMEs.

H2b. Dynamic capabilities significantly positively affect the relational collaboration of born-digital SMEs.

D. Collaborative Innovation and Born-Digital SMEs Performance

A few decades ago, innovation was perceived as a proprietary process of a single firm. Entrepreneurial activities and product innovation were believed to occur within a small business environment or a single department of a larger corporation. Even in digital technology-intensive firms, a predominant strategy was innovating internally [90]. The modern-day innovation

processes are altered by increasingly open, complex, and dynamic environments [91]. Innovation strategy has shifted from a unitized approach to technological collaboration that extends across firms, industries, and markets [87], [91]. Cross-domain collaboration has been established as the source of innovation and knowledge development [92].

This intrafirm collaboration and knowledge flow have been described as an “open innovation” [21]. Although the open innovation paradigm research is accelerating exponentially, researchers concentrate on some distinct areas [35], whereas other contexts, such as SMEs, are considered underdeveloped [75]. Open innovation is considered multilayered, so it is essential to understand the differences in each part and its unique antecedents. Previous literature has established individual, organizational, interorganizational, and industry/regional analysis [35]. However, the literature on the open innovation concept in born-digital SMEs is scarce and fragmented [12], [36], [68]. For this reason, areas such as the nature of collaborative innovation in SMEs have been developed sparingly. This has been documented and reasoned to be because of the different ways that SMEs and large firms benefit from open innovation.

Open innovation is based on a collaborative approach. It has become evident that combining a firm’s internal and external resources to interact with external opportunities and actors enhances innovation performance and helps acquire new knowledge and manage risks [18], [20]. Despite this, many SMEs can still not develop effective and long-term collaborative innovation practices.

According to Cui et al. [39], collaborative innovation is unique in the interfirm collaboration category since codevelopment is operationalized through partnering, cocreating, and sharing. Therefore, we operationalized collaborative innovation through partnership networks and relational collaboration to further develop this research.

Furthermore, born-digital SMEs are documented as benefiting from participating in partnership networks and bringing an immediate boost in internal and external capabilities by hiring senior executives from the market [4], [6], [93]. Researchers are concentrating on some distinct areas of the open innovation paradigm [35]; as a result, the literature on the open innovation concept in SMEs is scarce, underdeveloped, and fragmented [12], [24], [36]. For this reason, areas such as collaborative innovation’s nature in SME performance are still considered controversial [68]. Prior research on collaborative innovation focused on the interaction between different firms and other stakeholders [37], [38], stating that the current understanding of the collaborative innovation field is unique at the interorganizational level [39]. Yström and Agogué [38] construe collaborative innovation to mean innovation activities expressed through the involvement of several actors with a collective purpose. As a result, collaborative innovation grants access to an external knowledge base and assures joint efforts to develop new products and services.

Monaghan et al. [6] argue that born-digital SMEs do not necessarily perceive external relationships and partnership networks as a priority. This argument was built on proposing that Johanson and Vahlne [58, p. 1411] view current markets as “networks of

relationships in which firms are linked to each other in various, complex and, to a considerable extent, invisible patterns.” Instead, Monaghan et al. [6] build their case on digitalization and born-digitals’ capacity of engaging with stakeholders quicker than ever, decreasing digital technology innovation barriers to entry and sharing innovations directly.

In contrast, Vadana et al. [4] argue that born-digital SMEs could only achieve superior performance with the internal capabilities necessary for experimentation, rapid change, and learning [70] and the collective knowledge needed to seize the external benefits of relational collaboration and partnership networks. While these born-digitals have abundant digital design capital and technical competencies granting them digital options, some argue that networks seem to be a nonpriority for them [60].

Therefore, collaborative innovation is likely to mediate the relationship between dynamic capabilities and born-digital SME performance because it enables these firms to leverage their internal and external resources to create new products and services that meet the needs of their customers. By combining their dynamic capabilities with collaborative innovation, born-digital SMEs can develop new products and services more quickly and effectively, enhancing their market performance.

Based on the above, we formulate the following hypotheses.

H3a. Partnership networks have a significant positive effect on born-digital SMEs’ performance.

H3b. Relational collaboration has a significant positive effect on born-digital SMEs’ performance.

Experienced top management teams are a significant factor in sustaining SMEs’ superior performance as “dynamic capabilities reside in large measure with the enterprise’s top management team” [64, p. 1346]. However, dynamic capabilities have an indirect effect on firm performance through functional competence, such as open or collaborative innovations in our case [57], [59], [66]. Teece [64] argues that firm performance depends on specific assets and functional capabilities, which are affected by dynamic capabilities. Further, Teece [19] and Bogers et al. [63] explicitly linked dynamic capabilities and open innovation as functional capability that directly impacts a firm superior performance. They argue that successful open innovation rests on sensing, seizing, and reconfiguring external and internal sources of knowledge. Thus, open innovation will expand and orchestrate the number of partners that are complementors in knowledge or technology [19]. Besides, firms practicing open innovation must redesign and transform organizations to integrate external knowledge, which may course cultural change and shift in organizational relational patterns. As a result, collaborative innovation, which is a composite of open innovation, mediates the effect of dynamic capabilities on superior firm performance [57]. In line with this reasoning, we expect the following.

H4. Collaborative innovation mediates the relationship between dynamic capabilities and born-digital SMEs’ performance.

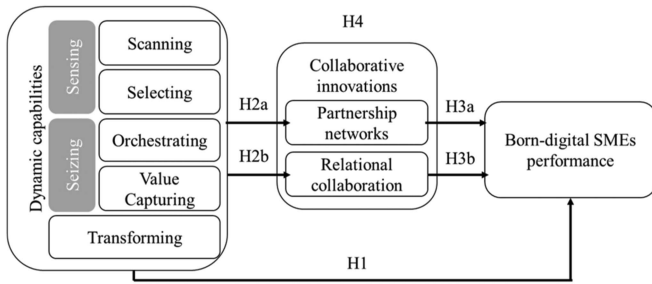


Fig. 1. Research model.

In summary, we provide a research model (see Fig. 1), which illustrates the hypothetical relationships between dynamic capabilities, collaborative innovation, and born-digital SMEs' performance. Next, we discuss the research methodology and provide data analysis and the four hypotheses testing results.

III. METHODOLOGY

A. Sampling Strategy

The study's sampling frame was born-digital SMEs operating in ICT, health services and medicine, and financial services industries based in USA, Sweden, and Lithuania. The born-digital SMEs sample was selected by identifying: 1) SMEs with solely digital products and services (e.g., digital platforms, SaaS, AI, etc.), operating on and leveraging digital infrastructure (e.g., cloud computing, digital data centers, etc.) [6], [33]; 2) SMEs having a digital business model and creating business offering online (e.g., selling online) [6], [33]; and 3) SMEs satisfying the first two criteria from the inception [4], [6].

The sample of respondents represents industry sectors characterized by high innovation intensity—ICT, health services and medicine, and financial services industries. Information and communications technology were the largest segment (63.4%), followed by health services and medicine (26.1%) and financial services (10.4%). Since dynamic capabilities are associated mainly with the organization's top management [55], primarily, business entity founders (43.3%) and C-Suite (50.7%) representatives were invited to fill in the questionnaires.

Finally, country innovativeness level was used as a control variable to test how different environments manifest in SMEs as a performance measure. Three countries were selected: The United States of America (represented by the state of California), Sweden, and Lithuania. According to the Global Innovation Index 2021 (GII) [94] and European Innovation Scoreboard 2021 (EIS) [43], Sweden (GII—ranked 2nd; EIS—ranked 1st) and the USA (state of California) (GII—ranked 3rd) are regarded as innovation leaders, while Lithuania is a moderate innovator (GII—ranked 40th; EIS—moderate innovator).

The sample size was determined by the number of SMEs in the country under investigation. To even out the demographics, we decided to use the state of California to represent the United States of America as a high-innovativeness economy. The state of California, with around 4 million SMEs [95], represented 41.7% of the sample, and Sweden, with 739 686 SMEs [96],

represented 30.7%. Lithuania, with 198 113 [97], accounted for 27.5% of the total representation (for the summary of the sample characteristics, see Table II).

The empirical research is based on 268 complete viable responses (see Table II) from born-digital SMEs operating in ICT, health services and medicine, and financial services industries, retrieved from the PitchBook global database.

B. Data Collection Procedures

Data were collected between September 2020 and January 2021 through an online survey of born-digital SMEs using SurveyMonkey as the questionnaire tool. The contacts of the SMEs were retrieved from the PitchBook global database with corresponding selection criteria of the relevant country SMEs as it is described in Section III-A. Using the criteria, we validated that the firms were born-digital SMEs from USA, Sweden, and Lithuania. Fig. 2 represents the research method diagram.

We undertook a number of procedures to ensure response quality depicted in Fig. 2. Out of the 40 698 collected email addresses, 30 172 were inactive or did not match our sample characteristics. Thus, after initial verification of the contacts, the invitations to participate in the survey were emailed to 10 526 SMEs with a comprehensive introduction, anonymity declaration, and a consent-to-proceed question. The introduction also indicated that there are no known risks related to participation, and no direct benefits from participation are expected for the respondent. Follow-up emails with additional enquiries were sent to improve the response rate. As a result, we received 1170 responses total (11.12% response rate we consider sufficient in online or web-based surveys with no monetary rewards [98]), out of which 856 were partially complete.

Furthermore, we removed 46 out of 314 complete responses because the respondents did not match some of the selection criteria, e.g., exceeded the number of 249 employees, provided non-born-digital technology-based answers for core products, and did not identify as part of the top management team. Each SME received only one questionnaire to complete. To counter the potential common method variance, the dependent and independent variable questions have been distributed separately in different parts of the questionnaire.

The questionnaire was designed to collect information about dynamic capabilities, collaborative innovation, performance of the born-digital SME, and general information about the SME, such as core product and service, infrastructure, size, age, and industry. Thus, we have ensured the response quality by:

- 1) applying well-defined sampling strategy;
- 2) creating a robust data collection process when we have eliminated contacts and responses not matching the sampling strategy;
- 3) explaining the essence of the study in the introductory email and testing the questionnaire language and comprehensiveness of the questions to the target group;
- 4) verification of the residence demographics partially by matching responses with demographic data provided in Pitchbook database and partially because the study was not offering monetary incentives.

TABLE II
DESCRIPTIVE CHARACTERISTICS OF THE SAMPLE

Industry	N	%	Participant country	N	%	Role in firm	N	%
Information and communications technology	170	63.4	USA	115	42.9	Founders	116	43.3
Health services and medicine	70	26.1	Sweden	82	30.6	C-Suite	136	50.7
Financial services	28	10.4	Lithuania	71	26.5	Top-level managers	16	6.0
TOTAL	268	100	TOTAL	268	100	TOTAL	268	100
Firm size	N	%	Firm age	N	%			
< 10 employees	113	42.2	< 3 years	69	25.7			
10-19 employees	64	23.9	3–5 years	75	28.0			
20-49 employees	57	21.3	6–10 years	72	26.9			
50-249 employees	34	12.7	> 10 years	52	19.4			
TOTAL	268	100	TOTAL	268	100			

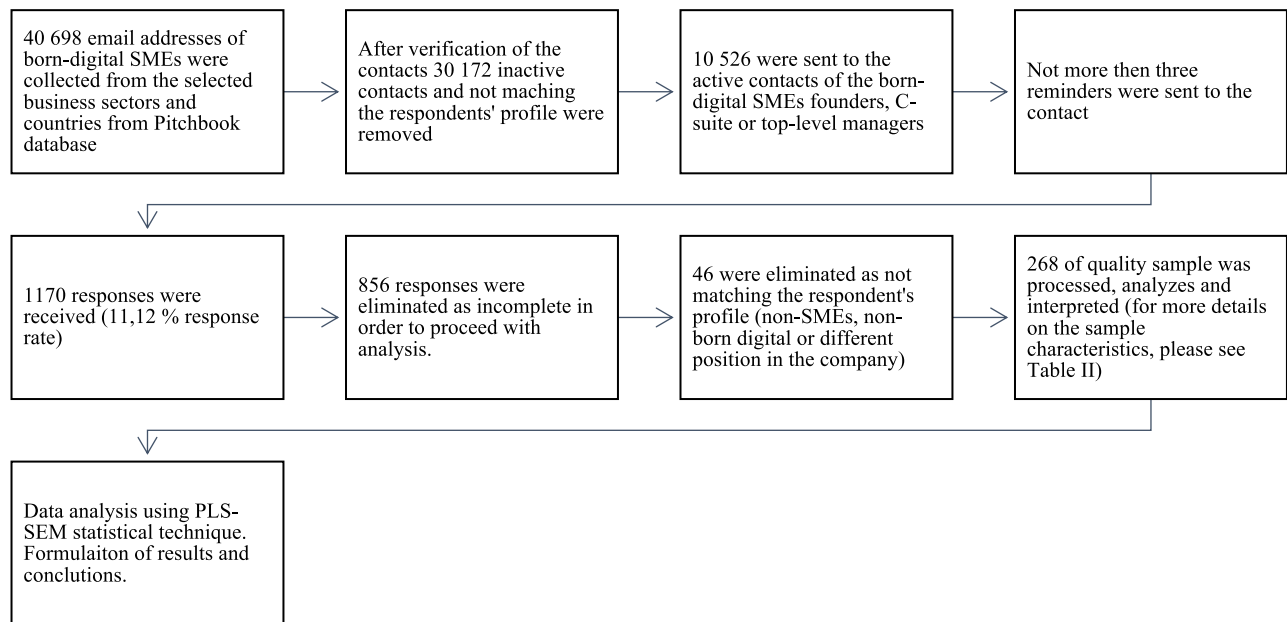


Fig. 2. Research method diagram.

Thus, participation in the survey was of free choice and no ultimate motive to falsify data.

C. Measurement of Constructs

A survey questionnaire was designed to test the conceptual model comprising dynamic capabilities, collaborative innovation, and performance of born-digital SME constructs. All construct indicators (38 items) were adapted from previously validated studies [59]. We assumed that validated scales in SME studies could measure dynamic capabilities, collaborative innovations, and performance constructs. However, due to the unique context of born-digital SMEs, the manifestation and interrelation of the constructs can be different from regular SMEs [54]. Adapting validated scales enabled us to meet the aim of the study to investigate the impact of the dynamic

capabilities of born-digital SMEs on their performance when collaborative innovations mediate it. Put differently, validated scales enabled us to assess the unique manifestation and interrelations among the established concepts. For example, we measured dynamic capabilities as a second-order construct with five subdimensions (scanning, selecting, orchestrating, value-capture, and transforming) (28 items) adapted from [19], [26], and [59]. Specifically, the “scanning” subdimension was operationalized through five items; e.g., “we assess trends of competitors’ performance regularly.” The “selecting” subdimension is measured with seven items based on such questions as “we proceed with product/service innovation based on the analysis of relevant external and internal data.” Six items measured the “orchestrating” subdimension through questions such as “our employees propose ideas for product/service innovation regularly.” The “value-capture” subdimension was measured through four

items; e.g., “we can develop a reliable business model for an innovative product/service”; and, finally, the “transforming” subdimension was operationalized through six items, e.g., “We reconfigure our organizational design to match business scale-up needs regularly.”

The collaborative innovation was constructed as a second-order construct composed of partnership networks and relational collaboration. Three items were measured in the “Partnership networks” variable: 1) “we collaborate with an extensive network of firms, academic institutions, associations, etc.”; 2) “to acquire a know-how/technology on innovative products/services, our organization cooperates with knowledge institutions, such as universities, faculties, institutes, and laboratories”; and 3) “to acquire know-how/technology on an innovative product/service, our employees cooperate with researchers from various knowledge institutions.” These three items were adapted from [35], [40], [41], and [83]. Further, the “relational collaboration” variable was measured by two items: 1) “we often cocreate an innovative product/service with different partners”; and 2) “we are sharing the cocreated value of an innovative product/service with involved partners.” These two items were adapted from [20], [82], [85], [86], [87].

Additionally, the performance as a first-order construct was measured by five items: SME’s growth compared to primary competitors, scaling performance locally and internationally, ability to attract investment, and growth of SME’s value. Firm-level performance has been identified as a consequence of dynamic capabilities manifested through such outcomes as financial or competitive firm accomplishments [62]. The survey questions and procedures were conducted considering the risk of common method bias since both the independent and dependent variable sources were the same [99]. However, these common method biases of self-reporting have been deemed misconceptions in recent years. As Conway and Lance [99, p. 327] explain: “the widespread belief that common method bias serves to inflate common method correlations as compared to their true-score counterparts is substantially a myth.” In addition, the common method bias of self-reporting perceived firm performance can be justified by the overall survey design since the respondents were informed about the anonymity of the data collected, thus, nullifying any incentives to inflate the scores. Furthermore, in Section IV, we provide extensive construct validity evidence, which includes factor structure and internal consistency reliability [99], [100], [101].

The items were measured using seven-point Likert scales, with options extending from 1 (“totally disagree”) to 7 (totally agree”). According to Weijters et al. [102], choosing the optimal number of scale gradation comes with a tradeoff “between maximizing the potential information transmission versus minimizing respondent demands.” They suggest that constructing 7-point rather than 5-point scales might be less problematic in cases where the respondents are considered to have high cognitive abilities and experience related to the main theme of the questionnaire.

The survey included questions about utilizing digital technologies, R&D intensity, size and age of the SME, relation to the technology, and the level of innovativeness of the SME’s country.

These have been used as control variables when validating and testing the conceptual model.

Within this article, we considered dynamic capabilities to be a second-order reflective-formative construct composed of five first-order reflective constructs (scanning, selecting, orchestrating, value-capture, and transforming). We utilized a two-stage model approach to estimate and assess dynamic capabilities as a second-order reflective-formative construct and to get the latent variable scores for lower order factors, which would indicate the higher order component [103], [104].

IV. ANALYSIS AND RESULTS

In this section, we first show and illustrate the respondents’ demographic information, then demonstrate and present the measurement and hypotheses testing. We have applied statistical techniques of partial least squares structural equations modeling (PLS-SEM). The PLS-SEM technique is relevant in our case because we analyze relatively small samples in Lithuania, Sweden, and the U.S. (see Table II). Therefore, PLS is more appropriate when the number of observations is around 250; in our case, we analyze 268 [105]. Furthermore, PLS works with nonparametric tests, which enables data nonnormality issues to be addressed [106]. We used Smart PLS v.3 software to analyze the data.

A. Measurement Model, Validity, and Internal Reliability

We computed the factor loadings and examined the composite reliability (CR) values to assess the internal consistency and validity of the items and constructs used in the model. The factor loading values were all above the threshold value of 0.70, as recommended by Hair et al. [107]. In addition, we calculated Cronbach’s alpha to assess the internal consistency of the items and the internal reliability of latent constructs. In our construct reliability evaluation, we examined CR. The recommended value for CR is 0.70 or higher [107]. In this article, the CR values above the recommended value were 0.845 as the lowest and 0.918 as the highest (see Table III). Next, we assess the discriminant validity to identify any interaction or relationship between the items (indicators) and the constructs. In other words, with this test, the researcher’s goal is to show that the items used to measure a construct indeed captured the intended construct. This means that discriminant validity, if established, indicates that the construct was not captured by other measures [108]. All the constructs in the model met the threshold values.

The correlation matrix (see Table IV) depicts that the dependent variable (performance) has a statistically significant positive correlation with dynamic capabilities, partnership networks, and relational collaboration. Thus, positive relations are expected to be revealed in the validated model.

As the model’s dependent variable (performance of digital SMEs) was predicted by multiple independent variables, such as dynamic capabilities and collaborative innovation, there might be a multicollinearity issue. To establish that our data and analyses are free of multicollinearity, we computed the variance inflation factor (VIF) values. Hair [109] recommends using the value of 5.0 as the cut-off value to determine that the data are free of multicollinearity issues. This article establishes that

TABLE III
DESCRIPTIVE STATISTICS

Concepts	Items	Loadings	Mean	Std.	Alpha	CR	AVE
Performance	PER1	0.72	4.907	1.667	0.772	0.845	0.522
	PER2	0.71	4.216	1.943			
	PER3	0.74	4.847	1.477			
	PER4	0.70	4.276	1.797			
	PER5	0.76	5.317	1.382			
Dynamic capabilities	SCA1	0.81	5.914	1.189	0.889	0.918	0.693
	SCA2	0.77	5.642	1.289			
	SCA3	0.77	5.929	1.071			
	SCA4	0.73	5.164	1.480			
	SCA5	0.80	5.414	1.250			
	SEL1	0.85	5.701	1.228			
	SEL2	0.78	5.780	1.216			
	SEL3	0.83	5.541	1.259			
	SEL4	0.82	5.601	1.231			
	SEL5	0.74	5.690	1.303			
	SEL6	0.79	5.881	1.182			
	SEL7	0.80	5.586	1.208			
	ORC1	0.83	5.772	1.186			
	ORC2	0.84	5.649	1.256			
	ORC3	0.81	5.496	1.340			
	ORC4	0.88	5.530	1.244			
	ORC5	0.84	5.940	1.115			
	ORC6	0.81	6.183	1.019			
	VAL1	0.89	5.575	1.260			
	VAL2	0.86	5.888	1.229			
	VAL3	0.86	5.593	1.334			
	VAL4	0.73	5.459	1.367			
	TRA1	0.76	5.888	1.157			
	TRA2	0.80	5.787	1.268			
	TRA3	0.87	5.407	1.265			
	TRA4	0.85	5.276	1.315			
	TRA5	0.86	5.493	1.268			
	TRA6	0.76	5.765	1.207			
Collaborative innovation	PAR1	0.72	4.646	1.912	0.846	0.905	0.762
	PAR2	0.71	4.597	1.871			
	PAR3	0.74	4.93	1.60			
	REL1	0.70	4.795	1.611	0.816	0.916	0.845
	REL2	0.76	4.944	1.531			

TABLE IV
CORRELATION MATRIX

	1	2	3	4
Performance	1			
Dynamic capabilities	0.399**	1		
Partnership networks	0.132*	0.454**	1	
Relational collaboration	0.311**	0.427**	0.387**	1

Note: * $p < 0.05$; ** $p < 0.01$; $N = 268$.

multicollinearity was not an issue in our data, as the lowest VIF was 1.420 and the highest was 4.089.

B. Hypotheses Testing

The structural equation modeling approach was used to assess and investigate the path relationships proposed in the conceptual model (see Fig. 3.).

The results indicated that a variance of 31% explained the performance of born-digital SMEs. Moreover, the collaborative innovation—partnership network—was explained by a variance of 22% and a variance of 18% explained relational collaboration. H1 was supported as the results showed that the path between digital SMEs' dynamic capabilities and performance is significant ($\beta = 0.35$, $t = 5.372$, $p < 0.001$). Moreover, the results showed that dynamic capabilities had positive and significant effects on both partnership networks ($\beta = 0.46$, $t = 9.049$, $p < 0.001$) and relational collaboration ($\beta = 0.41$, $t = 7.333$, $p < 0.001$). Both H2a and H2b were supported.

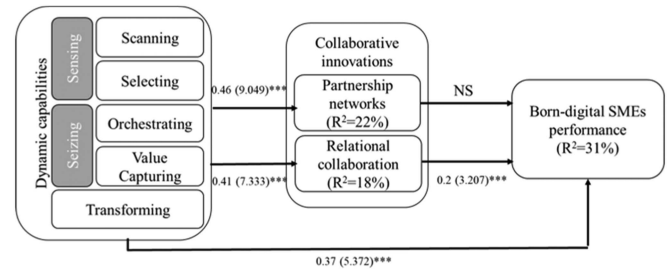


Fig. 3. Structural equation modeling.

The path between partnership networks and the performance of digital SMEs was not statistically significant. We rejected the H3a hypotheses. However, the path between relational collaboration and to the performance of digital SMEs was found to be significant ($\beta = 0.20$, $t = 3.207$, $p < 0.001$). The model supported H3b.

We conducted a mediation analysis to determine if collaborative innovation, which includes partnership networks and relational collaboration, plays a role in connecting dynamic capabilities with the performance of born-digital SMEs. The mediation analysis results showed that the path between the dynamic capabilities and to the performance of digital SMEs was significantly mediated through relational collaboration ($\beta = 0.08$, $t = 2.692$, $p < 0.005$). Thus, H4 was partially approved.

V. DISCUSSION AND CONCLUSION

Our study investigated how dynamic capabilities and collaborative innovation operationalized as partnership networks and relational collaboration affects born-digital SME performance. We focused solely on born-digital SMEs from three sectors: information and communication technology, health services and medicine, and financial services. So far, it has been widely discussed that, while innovating, SMEs need to rely on external resources accessed through partnership networks due to their size. Our article suggests that an extensive partnership network, which is perceived as a “loosely-coupled systems” [84], is not sufficient to enhance born-digital SME performance. The results of our study show that born-digital SMEs' superior performance rests on relational collaboration rather than solely on extensive partnership networks. This extends previous literature, which stated that successful asset base expansion depends on the “type of partnership and choice on collaboration mode” [24, p. 11]. Thus, digital-born SMEs need to have strong dynamic capabilities that help them do develop both: partnership networks and relational collaboration that intertwine personal and organizational relationships for the cocreation of digital innovations.

We demonstrate that dynamic capabilities significantly enhance collaborative innovation by fostering partnership networks and supporting relational collaboration. Even though dynamic capabilities and partnership networks do not significantly affect born-digital SME performance, dynamic capabilities and relational collaboration explain the superior performance of born-digital SMEs with a substantial variance of 31%. Dynamic capabilities by themselves have a less

significant impact on born-digital SMEs' performance, but this relationship is improved through the mechanism or mediation of collaborative innovation.

Collaborative innovation scholars researched how firms complement their tangible and intangible assets through technology exploration and scouting while trying to reduce innovation and transaction costs [24], [79], [80], [110]. Our study assessed the antecedents of collaborative innovations by taking an intrafirm perspective by deploying the dynamic capabilities view. Ultimately, we provided empirical evidence that the dynamic capabilities of born-digital SMEs ensure advanced collaborative innovation by supporting partnership networks and relational collaboration. None of the control variables in the model (innovativeness of the country, size of the SME, and age of the SME) was proven to have any significance. Our research generates a unique explanation of how born-digital SMEs' dynamic capabilities help to seize the benefits of collaborative innovation and to strengthen the overall performance of born-digital SMEs. In other words, we showed that dynamic capabilities are the driving force behind collaborative innovation and advancing born-digital SMEs' performance.

Previous literature has documented that technology exploration as part of collaborative innovation plays the main part in acquiring tangible and intangible assets needed for SMEs performance [24]. We confirm that dynamic capabilities, responsible for integrating, realigning, and reconfiguring resources critical to strategic change, positively affect collaborative innovation by fostering extensive partnership networks and exchange of knowledge with multiple cross-sectorial partners by supporting relational collaboration practices of cocreation of innovative products/services and sharing the cocreated value with relevant partners. In other words, the stronger the born-digital SME's dynamic capabilities, the more comprehensively it can seize the benefits created by partnership networks and relational collaboration.

Additionally, we extended previous literature on antecedents of open innovation in the context of SMEs and contributed to the latest scientific findings on collaborative alliances or, as we provide, relational collaboration [75]. Strong dynamic capabilities help define the core competence of digital-born SMEs and provide them with the ability to benefit from relational collaboration and partnerships, which also "reduces the risks connected to partners' opportunism" [111, p. 337].

The study's findings extend the existing MoT and innovation management literature in three ways. First, we bridge dynamic capabilities and collaborative innovation frameworks. Scholars have been researching dynamic capabilities as a means to digitally transform incumbent firms in traditional industries [23]. In contrast, we showed that dynamic capabilities are relevant to the born-digital SME context. Second, prior research on collaborative innovation focused on the interaction between different firms and other stakeholders [37], [38], stating that the current understanding of the collaborative innovation field is unique at the interfirm level [39]. We focused on the organizational level of collaborative innovation and, thus, the role of dynamic capabilities in advancing collaborative innovation activities through partnership networks and relational

collaboration. Our study provides empirical evidence that: 1) dynamic capabilities have a significant positive impact on both partnership networks and relational collaboration, thus, collaborative innovation; 2) only relational collaboration has a significant positive impact on born-digital SME performance; 3) finally, the relationship between dynamic capabilities and born-digital SME performance is moderated by collaborative innovations in particular relational collaborations. Third, in prior strategic management research, having an open strategy has been recognized as a source of dynamic capabilities in the past research [55], [112]. We suggested an opposite view, where dynamic capabilities are the antecedents of collaborative innovation. In addition, our study advances the current understanding of collaborative innovation by exposing the limited role of partnership networks in born-digital SMEs' superior performance and spotlighting the significance of relational collaboration—cocreating innovative products and services and sharing the cocreated value among the relevant partners.

This study has several implications for business practitioners. First, the research findings provide born-digital SME founders with new knowledge on collaborative innovation antecedents. This study shows that the prerequisites for improving born-digital SME performance are dynamic capabilities responsible for seizing the value from relational collaboration, such as cocreating innovative products/services and sharing the cocreated value. Without adequate dynamic capabilities, born-digital SMEs cannot benefit from collaborative innovation activities that are the primary stimulus for the superior performance of these firms. For this reason, born-digital SMEs should consistently invest their time and effort into building dynamic capabilities. Second, more is needed to cultivate extensive partnership networks. It is relational collaborations that significantly impact born-digital SMEs' performance. Thus, deeper relationships between the partners are needed to cocreate innovative digital products and services.

We look forward to qualitative research aiming at more in-depth analyses that can more explicitly explain the context and conditions of the causal links between dynamic capabilities, collaborative innovations, and born-digital SMEs' performance identified by our current study.

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